Lock-In Amplifier Experiment Lab Notebook[†]

Pritam Ghosh

March 4, 2025

1 Abstract

The Lock-In Amplifier (LIA) is a powerful instrument used to detect weak signals buried in noise by employing phase-sensitive detection. This experiment utilizes the Moku-Go device to explore the working principles of an LIA through various signal-processing scenarios. On the first day, different input signals—including pure sinusoids, multi-frequency signals, and square waves—are analyzed to understand the LIA's behavior in phase locking. The second day focuses on using the LIA for practical circuit measurements, such as voltage divider analysis and resistance extraction in noisy environments.

2 Aim

The aim of this experiment is to study the operation of a Lock-In Amplifier (LIA) using the Moku-Go device, focusing on its ability to detect weak signals in noisy environments. The specific objectives include:

- Understanding how the LIA locks onto a signal frequency and filters out noise.
- Investigating how different the reference frequency can be for the LIA performance.
- Exploring the LIA's performance on sinusoidal signals (multi frequency) and square wave.
- Using the LIA for practical circuit applications, such as resistance measurement in a voltage divider.

3 Apparatus

- Moku-Go device and software
- Breadboard
- Resistances

[†]Use Digital copy for the colored plots and diagrams.

4 Theory

A Lock-In Amplifier (LIA) is an essential instrument for detecting and measuring weak signals in the presence of noise. It achieves this by utilizing phase-sensitive detection, where a reference signal is used to extract the component of interest from a noisy background.

4.1 Principle of Operation

The core principle of an LIA is heterodyning, where an input signal S(t) is mixed with a reference signal R(t) of the same frequency. This results in two output components:

- A DC component (low-frequency term) that represents the desired signal.
- A high-frequency component, which is filtered out using a low-pass filter.

Mathematically, if the input signal is:

$$S(t) = A\cos(\omega t + \phi) \tag{1}$$

and the reference signal is:

$$R(t) = \cos(\omega t) \tag{2}$$

then their product is:

$$S(t).R(t) = A\cos(\omega t + \phi).\cos(\omega t) \tag{3}$$

Using the trigonometric identity:

$$\cos(A).\cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)] \tag{4}$$

we get:

$$S(t).R(t) = \frac{A}{2}[\cos(2\omega t + \phi) + \cos(\phi)] \tag{5}$$

The low-pass filter removes the high-frequency term, leaving only the DC component, which is proportional to $Acos(\phi)$, allowing accurate amplitude and phase measurements.

5 Diagram of the Lock-In Amplifier

The following diagram is shown in Moku-Go software.

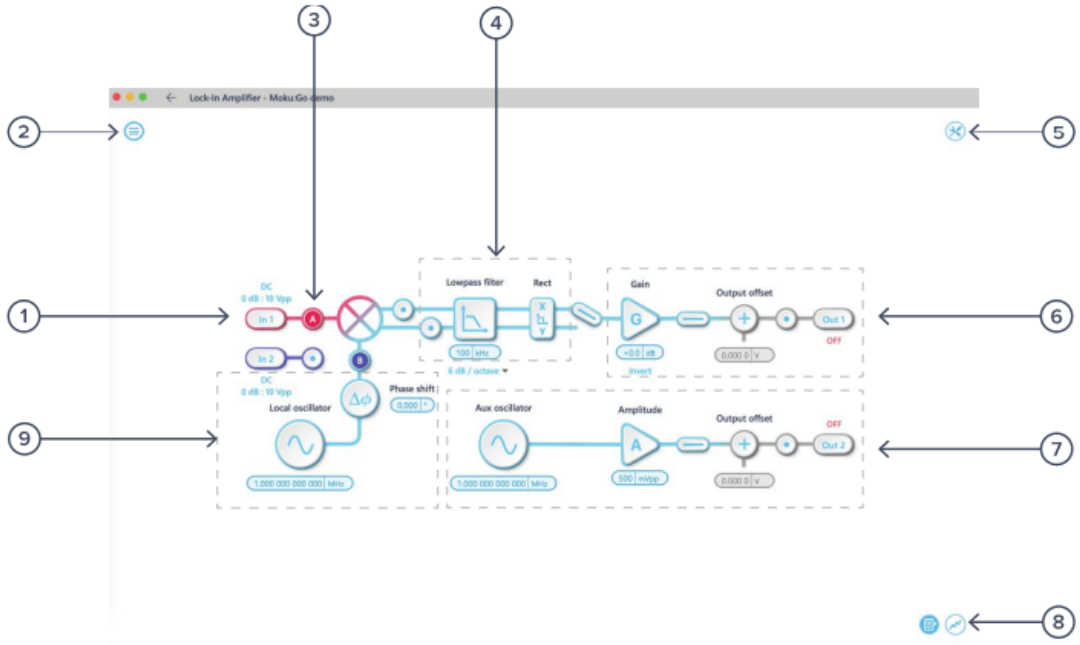


Figure 1: Lock-In Amplifier

6 Components

ID	Description	ID	Description
1	Input settings	6	Channel 1 output
2	Main menu	7	Channel 2 output
3	Probe point	8	Oscilloscope/Data Logger
4	Filter settings	9	Reference oscillator
5	Advanced configuration menu		

Figure 2: Key components Corresponding the Diagram

7 Experimental procedure

7.1 Day 1 (Understanding the LIA with Different Signals)

7.1.1 Case A: Single-Frequency Sinusoid with Matching Reference

When a sinusoidal signal with a single frequency is applied, and the LIA's internal reference frequency is set to match it, the LIA extracts the in-phase component. The phase-sensitive detection eliminates noise outside the reference frequency, producing a stable DC output proportional to the signal amplitude.

- Set up the Moku-Go in Lock-In Amplifier mode.

- Generate a single-frequency sinusoidal signal.
- Configure the LIA's reference frequency to match the signal frequency.
- Observe and record the LIA output amplitude.
- Compare the output with the expected theoretical value.

7.1.2 Case B: Single-Frequency Sinusoid with Slightly Different Reference

If the LIA's reference frequency is slightly different from the signal frequency, the output deviates from the expected value and the amplitude decreases.

- Generate a sinusoidal input signal
- Set the LIA reference frequency slightly different from the signal frequency.
- Gradually increase the frequency mismatch and note the point where the LIA fails to detect a signal whose amplitude deviates from the expected value.

7.1.3 Case C: Two Sinusoids – Low and High-Frequency Components

When an input signal contains two frequency components (one low and one high), we have to vary the low frequency component more low and the high frequency one more high and to note down where the output amplitude deviates from the expected value.

- Use the arbitrary waveform generator to create an input signal with two sinusoidal components: 1. One low-frequency component. 2. One high-frequency component.
- Decrease the frequency of the low frequency component keeping the high one fixed.
- Note down upto how much low value we get the expected results.
- Similarly do this for the high frequency component by increasing its frequency keeping the low one fixed.

7.1.4 Case D: Square Wave Input

A square wave consists of multiple harmonics, but an LIA detects only sinusoidal signals. It locks onto the fundamental frequency (first harmonic) while filtering out higher harmonics.

- Generate a square wave input using the waveform generator.
- Configure the LIA to lock onto the fundamental frequency of the square wave.
- Observe and record the extracted sinusoidal output from the LIA.
- Compare with the expected first harmonic of the square wave (Fourier analysis).
- Square wave is a fourier series of sinusoidal function, $f(t) = \frac{4A}{\pi n} \sum_{n=1,3,5...}^{\infty} \left(\sin \frac{2n\pi t}{L}\right)$ where n takes on only odd values, and for even values the output saturates to zero.

7.2 Day 2: Practical Circuit Applications of LIA

7.2.1 Voltage Divider Circuit (Measuring Resistance with LIA)

A voltage divider circuit allows resistance measurement by analyzing output voltage. The LIA can detect small signals across the low-value resistor (R_L).

- Construct a voltage divider circuit with a high-value resistor R_H and a low-value resistor R_L.
- Apply a sinusoidal voltage input to the circuit.
- Measure the output voltage across R_L.
- Measure the output V_{out} and calculate R_L from the equation $V_{\text{out}} = \frac{R_L}{R_H+R_L}V_{\text{in}}$.
- Detect R_L for more small values by connecting resistances parallel to R_L.

7.2.2 Voltage Divider with Noisy Input

When a noisy input signal is applied to the voltage divider, the LIA isolates the signal of interest while rejecting noise.

- We are provided with excel files consisting of noisy signals for different levels.
- We then upload these files in multi-instrument of Moku-Go software.
- Set the internal reference frequency to $1KHz$ and note the output value.

8 Experimental data

8.1 Day 1 experiments:

In the following experiments the phase difference between the reference and input signal is zero.

8.1.1 Case A:

Input Frequency(in kHz)	Demodulating Frequency(in kHz)	V_{in} (in volts)	V_{out} (in mV)
1	1	1	250.7
1	1	2	502
2	2	1	250.7
2	2	2	502
3	3	3	752.6

Table 1:

8.1.2 Case B:

Input Frequency(in kHz)	Demodulating Frequency(in kHz)	V_{in} (in volts)	V_{out} (in mV)
1	1.00005	1	250.7
1	1.0001	1	249.7
1	1.0002	1	245.5
1	1.0005	1	222.9
1	1.0008	1	193
1	1.0015	1	135.9
1	1.0025	1	90.76
1	1.005	1	47.78
1	1.01	1	24.73
1	1.02	1	12.67
1	1.04	1	6.412
1	1.08	1	3.74
1	1.2	1	1.679
1	2	1	0.610

Table 2:

8.1.3 Case C:

We take the input signal to be, $V_{\text{in}} = 0.4\sin(2\pi f_l t) + 0.5\sin(2\pi f_h t)$, where f_h is the high frequency of 10 kHz and f_l is the low frequency of 1 kHz.

First we will vary the low and high frequency and find out how far they can go low and high respectively without deviating from the expected output (which is $\frac{1}{4}$ of the input voltage).

Low Frequency(in kHz)	High Frequency(in kHz)	Demodulating freq(in kHz)	V_{out} (in mV)
1	10	10	125.3
1	10x10	10x10	125.2
1	10x100	10x100	124.7
1	10x200	10x200	123.2
1	10x250	10x250	122.1
1	10x300	10x300	121.1

Table 3: Varying high frequency keeping Low frequency fixed

High Frequency(in kHz)	Low Frequency(in kHz)	Demodulating freq(in kHz)	V_{out} (in mV)
10	1	1	100.1
10	1x0.9999	1x0.9999	99.54
10	1x0.9998	1x0.9998	98.01
10	1x0.99975	1x0.99975	96.94
10	1x0.9997	1x0.9997	95.41

Table 4: Varying low frequency keeping high frequency fixed

Now we will lock the LIA to the low frequency component and see if the amplitude of the high frequency component influence the output from the LIA.
We take the input signal to be -

$$V_{\text{in}} = 0.4\sin(2\pi f_l t) + A\sin(2\pi f_h t) \tag{6}$$

$f_l = 1$ KHz and $f_h = 10$ kHz , So the frequency of the low frequency wave is 1 KHz and that of the high frequency signal is 10kHz. We vary the amplitude of the high frequency signal by changing the values of A we get the following observations:

Demodulating Frequency(in kHz)	A (in volts)	V_{out} (in mV)
1	0.4	100.1
1	0.5	100.1
1	0.8	100.1
1	1	100.1

Table 5: Varying amplitude of high frequency component

8.1.4 Case D:

Even though a LIA generates only sine waveforms we can use it to demodulate a square wave because we can visualise a square wave as a superposition of many sinusoidal wave forms as we know from fourier series expansion. For a square wave we can expand it in fourier series to get :

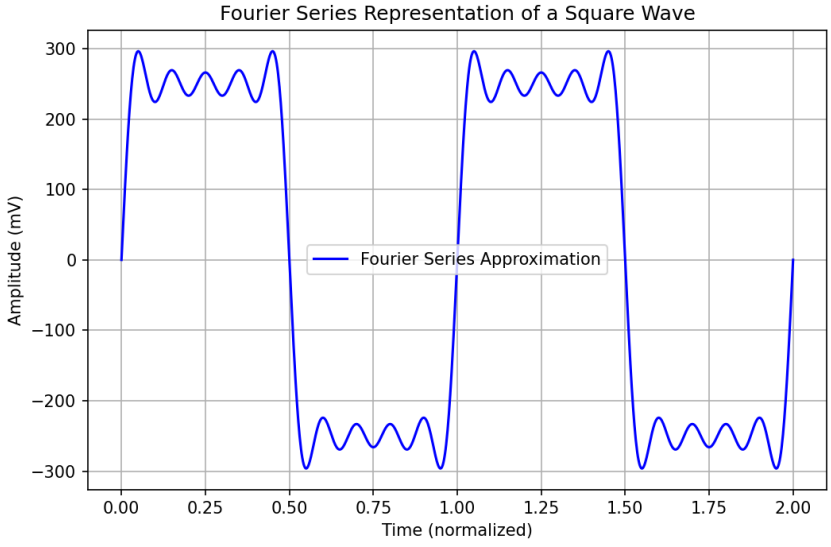
$$f(t) = \frac{4A}{\pi} \sum_{n=1,3,5...}^{\infty} \frac{1}{n} \left(\sin \frac{2n\pi t}{L} \right) \tag{7}$$

Demodulating Frequency(in kHz)	V_{out} (in mV)
1	318.3 (n = 1)
2	0.993
3	106.4 (n = 3)
4	0.534
5	63.97 (n = 5)
6	0.534
7	45.49 (n = 7)
8	0.534
9	35.57 (n = 9)

Table 6:

9 Fourier sum plot:

Using these values we can easily obtain-



9.1 Day 2 experiments :

9.1.1 Task 1:

Circuit Diagram:

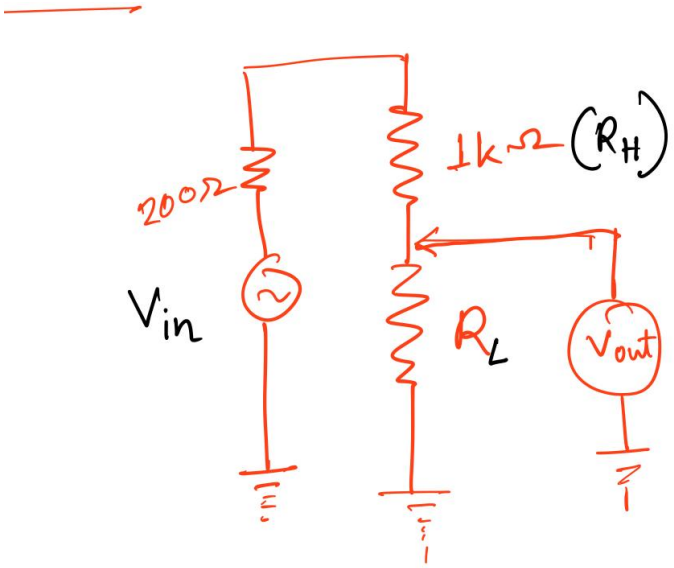


Figure 3: Voltage Divider circuit

We take an additional 200 ohms of resistance in series to the high resistance due to the internal resistance of the Moku-Go device. We now measure the voltage across the low resistance(R_L) $V_{\text{AB}} = V_{\text{out}}$. We then add another low resistance(R_L) parallel to the existing R_L to make the equivalent R_L lower and measure the output. We repeat this one more time.

Using KVL we get the following expression for R_L in terms of other known values :

$$R_{\text{AB}} = \frac{V_{\text{out}}(R_H + 200)}{V_{\text{in}} - V_{\text{out}}} \tag{8}$$

where $R_H = 1000$ ohms.

V_{in} (in volts)	V_{out} (in mv)	Resistance across AB	Calculated R_{AB}
1	2.137x4	R_L	10.35
2	4.275x4	R_L	10.35
3	6.412x4	R_L	10.35
4	8.396x4	R_L	10.16

Table 7:

V_{in} (in volts)	V_{out} (in mv)	Resistance across AB	Calculated R_{AB}
1	1.069x4	$R_L \| \| R_L$	5.15
2	2.137x4	$R_L \| \| R_L$	5.15
3	3.206x4	$R_L \| \| R_L$	5.15
4	4.275x4	$R_L \| \| R_L$	5.15

Table 8:

V_{in} (in volts)	V_{out} (in mv)	Resistance across AB	Calculated R_{AB}
1	0.610x4	$R_L \| \| R_L \| \| R_L$	2.94
2	1.679x4	$R_L \| \| R_L \| \| R_L$	4.04
3	2.137x4	$R_L \| \| R_L \| \| R_L$	3.43
4	3.206x4	$R_L \| \| R_L \| \| R_L$	3.86

Table 9:

9.1.2 Task 2:

Frequency = 1 kHz, V_{in} = 1 V

Noise percentage	V_{out} (in mv)
10	76.94
20	112.7
30	107.9
40	109.1
50	102.7
80	90.07

Table 10:

9.1.3 Task 3:

Here we generate input signal with the help of Moku Go and measure it with the help of the DSO, to find the minimum range of the DSO.

Table 11: Input output voltage

SF.	Moku Go input voltage (mV)	DSO output (mV)
1	1000	1200
2	500	500
3	600	420
4	700	720
5	600	620
6	400	420
7	600	420
8	200	220
9	300	220
10	100	150
11	50	100
12	70	Reminate

For Data:

Data Table:

For DMM

Finding the working frequency range of DMM, using Moku Go, we generate an input unique adapter's signal generator in Moku Go and then fit to DMM.
If $\mathbf{V_{in}}$ will be given as input to the DMM then the expected output will be $\mathbf{V_{out}}$.
$\frac{V_{in}}{2\sqrt{2}} = V_{out}$
Here we set the input voltage amplitude to be 1 volt. Then the output voltage should be:
$\frac{1}{2\sqrt{2}} = 0.326\nu$
Here we feed DMM frequency ranging in kHz and Hz to find its working range.

For kHz

Table 12: Frequency and amplitude

SF.	Moku Go input frequency (kHz)	DMM output (mV)
1	1	0.348
2	10	0.363
3	20	0.283
4	30	0.312
5	40	0.322
6	60	0.292
7	80	0.364
8	90	0.264
9	90	0.262
10	100	0.267
11	160	0.191
12	200	0.069

For lower range:

Table 13: Frequency and amplitude

SF.	Moku Go input frequency (Hz)	DMM output (mV)
1	1	0.042
2	2	0.374
3	3	0.455
4	5	0.432
5	6	0.422
6	7	0.386
7	8	0.378
8	9	0.271
9	10	0.268
10	20	0.302
11	80	0.363

10 Inferences

10.1 Day 1 experiments:

- In case A, As the demodulating frequency and the input frequency are same, we get exactly the one half of the amplitude of the input signal or one fourth the peak to peak voltage of the input signal.

- In case B, As the difference between the reference frequency and the input frequency increases, the output voltage drops rapidly. This occurs because a lock-in amplifier extracts only the signal that matches the demodulation frequency, while all other signals are attenuated. If input frequency is 1 kHz, for demodulating frequency of 1.0005 the output voltage deviates from the expected result.

- In case C, keeping low frequency fixed,a change in 250 times the high frequency i.e, 250x10 kHz = 2.5 MHz, the ouput deviates from the expected result. And keeping high frequency fixed, a change in 0.99975 times the low frequency, the output deviates from the expected output. We see that a very little change in low frequency the output changes drastically whereas for high frequency it changes very slowly.

  The change of the amplitude of the higher frequency signal has no effect on the output of the LIA. This happens because the demodulating frequency is set to the lower frequency signal and the higher frequency signal is attenuated anyways.

- In case D, from equation 7, we see that square waves can be expanded in superposition of sines waves, where n corresponds to only odd values. From the experimental data we see that for n = even values the ouput attenuates to zero. For odd values of n, we need to lock the LIA.

  n = 1 corresponds to the first harmonic and its peak value is greater than the expected value(i.e, $\frac{1}{4}$ of the amplitude of input voltage) because within the first

harmonic all the higher order harmonics lie which on superposition lowers its peak value.

10.2 Day 2 experiments:

- For experiment 1, we got the value of R_L as 10.35 ohm and when two and three R_L are connected parallel to each other, we got the equivalent value of R_L to be 5.15 and 3.57 respectively.
- For experiment 2, we are given excel files containing noise signals of different levels. Ideally LIA should be able to extract the signal from noise. So, for a reference frequency of 1 kHz we have to note down the output amplitude for different noise percentage.
- for experiment 3:
  – DSO provides accurate measurements down to 100 mV. Below this threshold, the voltage readings exhibit fluctuations.
  – ADMM provides accurate measurement in range of 80 Hz to 30 Khz after that the the value provided differs greatly from the analytical value.

11 Sources of error

- Instability in the signal source or measurement equipment can occur.
- Accurately generating and controlling noise levels can be challenging.
- The LIA may struggle to separate signals from extreme noise conditions.
- A phase mismatch between the reference and signal affects the output amplitude.
- The LIA's output can small be influenced by higher-frequency components.
- Choosing a reference that doesn't match the square wave's fundamental frequency can cause issues.
- Measurement accuracy can be disrupted by random voltage fluctuations.